

# The Economics of Semiconductor Yield

## What Process Node Competition Actually Measures

Michigan Journal of Economics

---

When news coverage describes SMIC producing a 7nm chip, the implicit frame is one of technical parity: China's leading fab has reached the same process node that TSMC uses. What that framing omits is the economic structure that converts a process capability into a competitive product. A fab that can demonstrate a 7nm chip is not the same as a fab that can produce 7nm chips at a cost and volume that makes them commercially viable. The variable that governs that difference is yield, and yield is among the least discussed but most consequential parameters in semiconductor economics.

This paper lays out the economic structure of semiconductor manufacturing yield, explains why yield improvement is non-trivial, and shows why a persistent yield gap between Chinese and Taiwanese fabs represents an economic disadvantage that raw process node comparisons cannot capture.

### *The Yield Equation*

Semiconductor yield is the fraction of chips on a processed wafer that function correctly and meet performance specifications. A wafer that begins production as a featureless silicon disk passes through several hundred processing steps before the individual chips are cut apart and tested. At each step, imperfections in materials, equipment, and process control can introduce defects. A defect that falls on a critical circuit element destroys that chip. The yield of a process is therefore a function of the defect density per unit area and the area of the individual chip die.

The standard industry approximation, often called Murphy's model, captures this relationship:

$$Y = \left(1 - \frac{A \cdot D}{N}\right)^N$$

where Y is yield, A is die area, D is defect density per unit area, and N is a factor reflecting the distribution of defects across the wafer.<sup>1</sup> The relationship has two immediately useful implications. First, yield falls as die area increases: a chip that occupies twice the silicon

area has a much lower yield at the same defect density. Second, yield improves as defect density falls through process maturation. A newly introduced process node has high defect density; as engineers identify and eliminate defect sources over months and years, defect density falls and yield rises toward an asymptote.

Cost per working chip follows directly:

$$\text{Cost per die} = \frac{\text{Cost per wafer}}{\text{Dies per wafer} \times \text{Yield}}$$

A 300mm wafer processed on TSMC's N5 node carries an estimated fabrication cost in the range of \$16,000 to \$17,000.<sup>2</sup> At a typical AI accelerator die size of roughly 800 square millimeters, a wafer holds approximately 58 dies at theoretical perfect packing. At a mature N5 yield of roughly 70 to 75 percent, that wafer produces 40 to 43 functional dies at a cost of roughly \$370 to \$425 each. A competing fab running the same die area at a 35 percent yield produces 20 functional dies from the same wafer area and must charge commensurately more per chip simply to recover fabrication cost, before any profit margin.

### ***Why EUV Changes the Economics***

Advanced logic chips at and below 7nm require patterning features smaller than a single exposure of 193nm deep ultraviolet (DUV) light can resolve. Two approaches exist for printing such features: EUV lithography, which uses 13.5nm light to print fine features in fewer steps, and multi-patterning with DUV, which approximates the same resolution through repeated exposures on successive masks.

The economic implications of this choice are substantial. A 7nm chip at TSMC requires approximately nine EUV lithography steps for its most critical layers. Achieving equivalent feature resolution with DUV multi-patterning requires approximately 34 steps.<sup>3</sup> Each additional patterning step introduces another opportunity for alignment error: the features printed in one exposure must align precisely with those from a previous exposure, with tolerances measured in single-digit nanometers. Misalignment between patterning steps creates edge placement error, which causes some circuits to fail. The probability of alignment error compounds multiplicatively across patterning steps.

This is the structural source of the yield disadvantage associated with DUV-based advanced node production. It is not primarily a matter of engineering skill or process maturity; it reflects the additional defect generation opportunities that multi-patterning introduces regardless of execution quality. A fab producing 7nm-class chips via DUV self-aligned quadruple patterning will have a higher defect density at comparable process maturity than

a fab using EUV for the same layers, because defect sources that EUV eliminates remain active under multi-patterning.

The capital cost differential compounds the operating disadvantage. Each EUV system from ASML costs approximately \$150 to \$200 million.<sup>4</sup> Each DUV immersion tool costs \$40 to \$60 million. But because DUV multi-patterning requires 3 to 4 times as many exposure steps as EUV for equivalent feature resolution, the effective throughput cost per wafer is higher under DUV despite the lower tool price. When tool amortization, consumables, and cycle time are included, industry estimates place the all-in wafer cost for DUV-based advanced node production at 40 to 50 percent above EUV-based equivalents at the same nominal process node.<sup>5</sup>

### ***The Learning Curve and Its Limits***

Process yield does improve over time. The semiconductor industry's well-documented learning curve shows that defect density falls as cumulative wafer volume increases, engineers accumulate process data, and equipment suppliers refine tool performance. A process node that launches at 30 percent yield might reach 70 percent yield within 18 to 24 months of volume production.

For a fab using DUV multi-patterning at advanced nodes, the learning curve applies but with a structural ceiling. As noted above, multi-patterning introduces defect sources that are absent under EUV. Some of these sources respond to process optimization; others are intrinsic to the physics of the approach. Industry observers estimated SMIC's 7nm yield at roughly 15 percent at initial production ramp in 2023, improving to below 50 percent by mid-2024.<sup>6</sup> TSMC's N7 process, which launched in 2018, reached mature yields approaching 80 percent within two years. That gap reflects not only process maturity differences but the structural yield ceiling imposed by multi-patterning's additional defect exposure.

The ceiling matters for understanding the competitive trajectory. It means that even as SMIC gains experience with its advanced node processes, the cost-per-chip convergence with TSMC is bounded. SMIC will not reach TSMC's yield at the same nominal node using DUV multi-patterning, because the defect sources that EUV eliminates will remain active. Full cost parity requires either access to EUV or a future in which China develops domestically capable extreme ultraviolet tools – a technical challenge that has occupied the world's leading optics firms for three decades.

### **Implications**

The economic analysis above has a direct bearing on the question of semiconductor self-sufficiency. A fab that can demonstrate a 7nm chip demonstrates process capability. It does not demonstrate the ability to supply that chip at commercial cost and volume. The difference between those two claims is yield, and yield at advanced DUV-based nodes in China is structurally lower than at TSMC's EUV-based equivalents by an amount that translates directly into cost per chip and effective supply capacity.

For AI accelerator production specifically, where individual chips cost hundreds of dollars and a competitive training cluster requires tens of thousands of chips, the yield-driven cost premium has material consequences for the scale of compute China's domestic industry can actually provision – a consideration that process node announcements, taken at face value, systematically obscure.

---

### **References**

<sup>1</sup> C.H. Stapper, "Modeling of Defects in Integrated Circuit Photolithographic Patterns," *IBM Journal of Research and Development* 28, no. 4 (1984): 461-475. The Murphy model is the standard industrial approximation; actual yield functions vary by process and defect distribution.

<sup>2</sup> IC Knowledge LLC, *Cost and Price Model for Semiconductor Manufacturing* (2023 edition). Wafer cost estimates for leading-edge nodes are proprietary and vary by source; the figures cited reflect commonly reported industry estimates for TSMC N5 as of 2023.

<sup>3</sup> EDN, "The Truth About SMIC's 7-nm Chip Fabrication Ordeal" (August 2022); TechInsights, "SMIC's N+1 Process Node Analysis" (2023). The 9-step EUV versus 34-step DUV comparison is consistent across multiple independent process engineering analyses of SMIC's multi-patterning approach and is derived from mask layer counts in TechInsights teardown work.

<sup>4</sup> ASML Holding N.V., *Annual Report 2022* (2023). Machine prices are periodically updated in ASML's public disclosures and investor presentations.

<sup>5</sup> Financial Times, "SMIC Plans to Use Existing Equipment to Mass-Produce 5nm Chips" (February 2024); Asia Times, "SMIC to Sell Huawei Costly, Inefficient 5nm Chips" (February 14, 2024), citing chip engineers who estimated DUV-based 5nm production costs as several times higher than equivalent EUV-based processes. The 40-50 percent cost premium estimate specifically for SMIC versus TSMC at comparable nodes reflects

analyst assessments from 2023-2024 cited in Reuters and Financial Times coverage of Huawei chip pricing.

<sup>6</sup> Reuters, cited in multiple trade press sources including EDN and TrendForce (2023-2024). SMIC does not publicly disclose yield rates; estimates range from conservative industry analyst figures to more pessimistic assessments by those with supply chain access.